NetraRakshak: Diabetic Retinopathy Screening with an Explainable CNN and a Clinician Sign-Off Workflow

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***Abstract*— We built NetraRakshak, a web application in which a technician uploads both fundus photographs of a patient, the software checks image quality, a convolutional network suggests a diabetic retinopathy (DR) grade, and an ophthalmologist must accept, change or reject that suggestion before a report can be issued. The grading model is EfficientNet-B0 fine-tuned on the 3,662 labelled images of the APTOS 2019 dataset. On a held-out test split of 524 images it graded 429 correctly (81.9%, 95% CI 78.3–84.9%), with a quadratic weighted kappa of 0.876. For referable DR (moderate or worse) it detected 168 of 205 cases (sensitivity 82.0%, CI 76.1–86.6%) and correctly cleared 308 of 319 non-referable cases (specificity 96.6%). Temperature scaling lowered the expected calibration error from 12.2% to 6.1%. We describe how the model was trained and evaluated, how Grad-CAM maps are produced, how the clinician workflow was tested, and the main threat to these numbers: APTOS has no patient identifiers, so the split could not be made per patient. The system has not been validated on patients.**

***Index Terms*— diabetic retinopathy, fundus photography, EfficientNet, Grad-CAM, calibration, clinical decision support**

I. Introduction

People with diabetes are advised to have their retina photographed regularly, because diabetic retinopathy usually causes no symptoms until it is advanced. Taking the photograph is quick; grading it needs a trained reader, and there are far fewer ophthalmologists than people who need screening. Several groups have shown that convolutional networks can grade fundus photographs with high accuracy [1–3].

Our project asked a narrower, practical question: what does a hospital need around such a model so that a doctor stays in charge of every result? We wrote the software to answer it and trained our own model to put inside it. This paper reports both parts, with enough detail for the results to be checked. Specifically, we (i) trained and calibrated a five-grade DR classifier and measured it on data it had not seen; (ii) added a rule that withholds a grade when the model is unsure; (iii) generated Grad-CAM maps for every graded image; and (iv) built a workflow in which only a password-confirmed clinician signature can turn a draft report into an approved one.

II. Related Work

Gulshan *et al.* [1] trained an Inception-v3 network on about 128,000 graded images and reported an area under the ROC curve of about 0.99 for referable DR. Ting *et al.* [2] validated a system across several ethnic groups and cameras. Ruamviboonsuk *et al.* [4] compared a deep learning system with graders in Thailand's national programme, and Abràmoff *et al.* [3] ran a pivotal trial of an autonomous system in primary care (sensitivity 87.2%, specificity 90.7%). Rajalakshmi *et al.* [5] studied AI grading of smartphone-based fundus images in India. Our model is far smaller in data and scope than these systems; we rely on public data (APTOS 2019 [6]) and on two standard tools, Grad-CAM [14] and temperature scaling [13].

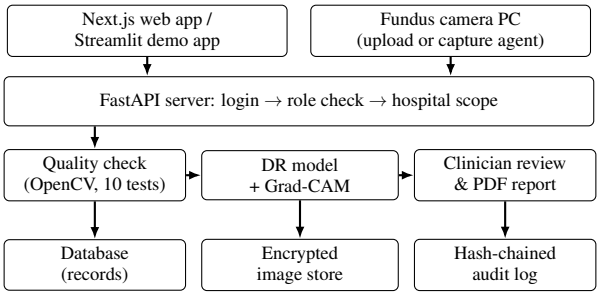


Figure 1: Components of NetraRakshak.

III. System Overview

Fig. 1 shows the parts. The web front end is written in Next.js; the server is FastAPI with a PostgreSQL database (SQLite during development); the model runs in PyTorch inside the server process. A second front end written in Streamlit drives the same server code and is the version prepared for public demonstration.

A screening proceeds as follows. Reception registers the patient and records consent for screening and, separately, for AI analysis. The technician uploads the right- and left-eye images. Ten quality tests run on each image (Table 1); a failed test returns "recapture required" and the reason. The model runs only on images that pass, and only if AI consent is on record. The ophthalmologist then sees the images, the model's probabilities and the Grad-CAM map, records a grade per eye, and signs by re-entering their password. A report generated before signing is watermarked "draft"; after signing it is marked "clinician approved" and the screening is locked. The model's output is stored in its own table and is never overwritten by the doctor's grade.

Table 1: Image Quality Tests (run before the model)

Test	What is measured
Resolution	shorter side ≥ 512 pixels
Retinal visibility	share of the frame covered by the retina
Field of view	distance of the retina's centre from the image centre
Fundus plausibility	red channel dominant inside the retina
Brightness	mean luminance inside the retina
Contrast	standard deviation of the green channel
Exposure	share of over- and under-exposed pixels
Blur	variance of the Laplacian
Focus	mean squared Sobel gradient
Artefacts	uneven illumination, large reflections

Table 2: Images per Grade in Each Split

Split	No DR	Mild	Moderate	Severe	PDR	Total
Train	1281	281	710	134	209	2615
Validation	251	43	156	34	39	523
Test	273	46	133	25	47	524

Each hospital's records are kept separate in every database query, staff have one of five roles, images and reports are encrypted before storage, and every view, upload, analysis and signature is written to an audit log in which each entry includes a SHA-256 hash of the previous one.

IV. Model Development

A. Data and Split

APTOS 2019 [6] contains 3,662 colour fundus images from Aravind Eye Hospital, each graded 0–4 on the International Clinical DR scale [8]. We used it under the Kaggle competition rules accepted by our account; we have not confirmed that it may be used in a commercial product. We split the images approximately 71/14/14 into training, validation and test sets with scikit-learn's `StratifiedGroupKFold` (random seed 42), stratified by grade (Table 2). The splitting code groups images by patient, but APTOS does not publish patient identifiers, so each image formed its own group. We return to this in Section VI.

B. Preprocessing

Each image is cropped to the circular retina, padded to a square and resized to 384×384. We then subtract a Gaussian-blurred copy, $I' = 4I - 4\,(G_\sigma * I) + 128$ with $\sigma = 384/30$, as Graham did in the 2015 Kaggle competition [9]; this evens out illumination differences between cameras. Pixels are scaled with the ImageNet mean and standard deviation. Inference uses the same function as training. During training, images are randomly flipped, rotated (up to $\pm 180^\circ$), rescaled (0.92–1.10) and given small brightness and contrast changes.

C. Network and Training

We used EfficientNet-B0 [10] from torchvision with ImageNet weights [11], replacing the classifier with dropout ($p = 0.3$) and a linear layer with five outputs (4.01 M parameters in total). The loss was cross-entropy with label smoothing 0.05 and class weights inversely proportional to class frequency, to counter the dominance of grade 0. We trained with AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}), a one-cycle learning-rate schedule, batch size 24 and mixed precision, for at most 25 epochs, keeping the checkpoint with the best validation quadratic weighted kappa [12] and stopping after six epochs without improvement.

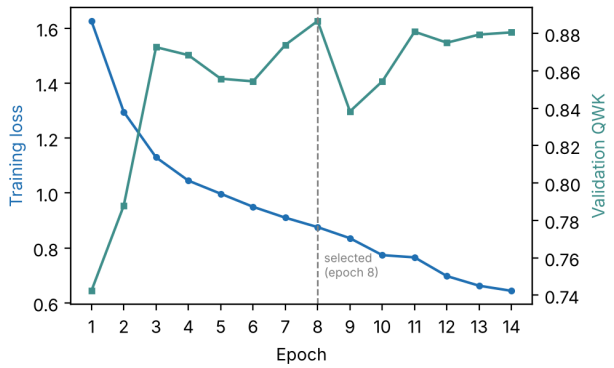


Figure 2: Training loss and validation quadratic weighted kappa per epoch (values logged by the training script).

Training ran in a Kaggle notebook on one NVIDIA GPU with PyTorch 2.11.0 and Python 3.13. It stopped after 14 epochs and kept epoch 8 (validation kappa 0.887). The first epoch took 244 s because it also preprocessed and cached the images; later epochs took about 24 s, for 9.3 minutes in total. Fig. 2 shows the logged training loss and validation kappa. The validation kappa peaks early and then fluctuates while the training loss keeps falling, a sign that longer training would mainly have fitted the training set more closely.

D. Calibration and Withholding Uncertain Grades

Softmax outputs of modern networks tend to be overconfident [13]. We fitted a single temperature T to the validation logits by minimising the negative log-likelihood (L-BFGS), obtaining $T = 0.771$. In use, the probabilities are $p_k = \mathrm{softmax}(z/T)_k$. The application shows the top grade only if its probability is at least 0.60 and the normalised entropy $-\sum_k p_k \log p_k / \log 5$ is at most 0.75; otherwise it shows "inconclusive — human review required". These thresholds are defaults and have not yet been tuned on the validation set.

E. Grad-CAM Implementation

For each analysed image we compute Grad-CAM [14] on the output of the last convolutional block of EfficientNet-B0 (`features[-1]`, a 12×12 feature map with 1,280 channels at our input size). A forward hook stores the activations A^k and a backward hook stores the gradients of the predicted class's logit y^c. The weight of each channel is the spatial mean of its gradient, $\alpha_k = \frac{1}{144} \sum_{i,j} \partial y^c / \partial A^k_{ij}$, and the map is $\mathrm{ReLU}(\sum_k \alpha_k A^k)$, divided by its maximum. The map is enlarged to 512×512 with bicubic interpolation, coloured with the JET colour map and blended with the photograph (42% map, 58% image). Both the overlay and the plain heatmap are stored, encrypted, alongside

Table 3: Test-Set Results (n = 524, one training run)

Measure	Value (95% CI)
Accuracy, 5 grades	81.9% (78.3–84.9)
Quadratic weighted kappa	0.876
Macro F1	0.670
Macro ROC-AUC (one-vs-rest)	0.917
Referable DR sensitivity	82.0% (76.1–86.6), 168/205
Referable DR specificity	96.6% (93.9–98.1), 308/319
Referable DR PPV / NPV	93.9% / 89.3%
Referable DR ROC-AUC	0.981
Calibration error (ECE), before / after	12.2% / 6.1%

Table 4: Results per Grade

Grade (test images)	Sens.	Spec.	Prec.	F1	AUC
No DR (273)	99.3%	96.8%	97.1%	0.982	0.998
Mild NPDR (46)	71.7%	93.1%	50.0%	0.589	0.935
Moderate NPDR (133)	64.7%	93.1%	76.1%	0.699	0.946
Severe NPDR (25)	44.0%	95.8%	34.4%	0.386	0.810
Proliferative DR (47)	59.6%	98.7%	82.4%	0.691	0.895

the result. The interface and the PDF label them as an explanation of the model's output, not as detected lesions.

V. Results

A. Test-Set Performance

All values below come from the 524 test images, which were not used for training, for choosing the checkpoint or for fitting the temperature. The training script computed them from the saved predictions and wrote them into the model's metadata file. Fig. 3 shows the confusion matrix. Its diagonal holds $271 + 33 + 86 + 11 + 28 = 429$ correct grades, so the accuracy is $429/524 = 81.9\%$. For the referable/non-referable decision, grades 2–4 are pooled: 168 of the 205 referable test images were predicted as grade 2 or higher (sensitivity 82.0%), and 308 of the 319 non-referable images as grade 0 or 1 (specificity 96.6%). Confidence intervals in Table 3 are 95% Wilson intervals. They come from a single training run; we did not repeat training with other seeds.

The model is reliable at recognising eyes without retinopathy (271 of 273). Most of its errors are between neighbouring grades: 26 moderate cases were called mild, and 12 severe cases were called moderate. Severe NPDR, with only 25 test images, is the weakest grade (11 correct; sensitivity CI 26.7–62.9%).

B. Speed

On a two-core 2.1 GHz Intel Xeon CPU with no GPU, one analysis including Grad-CAM took a median of 304 ms (90th percentile 331 ms over 10 runs), and the ten quality tests took about 73 ms.

C. Does the Workflow Work?

We tested the software at three levels. First, 30 automated tests check, among other things: that one hospital cannot read another's records; that each role is refused actions outside its permissions; that blurred, dark, low-resolution and non-fundus images are rejected; that no AI result is produced without consent, for a failed image, when no model is installed, or when the model file's SHA-256 does not match its metadata; that an uncertain prediction is shown as inconclusive; that signing fails without the attestation or with a wrong password; that a report is "draft" before signing and "clinician approved" after; and that the audit hash chain verifies. All 30 pass. Second, a scripted browser session completed the whole workflow (registration, upload of both eyes, quality check, analysis with this model, review, signature, PDF, dashboard) against the server running in production mode on PostgreSQL. Third, the same sequence was repeated through the Streamlit demo. Fig. 4 shows the review screen from the first front end.

VI. Threats to Validity

Leakage between splits. Because APTOS lacks patient identifiers, the two eyes of one person, or repeated photographs of the same eye, may sit in different splits. Near-duplicate images in APTOS have also been reported by Kaggle participants; we did not check for them. Both effects would make the test results look better than they are. A per-patient split on a dataset with identifiers, plus duplicate detection by perceptual hashing, is needed before the numbers can be trusted.

Single source. All images come from one hospital network. Performance on other cameras, populations, or undilated pupils is unknown; we have not evaluated on an external set such as IDRiD [7].

Single run. The intervals in Table 3 describe sampling of the test set only, not variation between training runs.

Inputs outside the training distribution. The quality tests check image properties, not whether the image is a retina the model has learned to grade. When we fed the model procedurally generated synthetic test images, it returned confident grades. An out-of-distribution check is needed.

Missed referable cases. At the default operating point, 37 of 205 referable eyes (18%) were graded non-referable. This is the main reason the software requires a clinician's signature on every case.

VII. Reproducibility

The model file used for all results has SHA-256 `d25443af...eaad9b`; the application refuses to load it if the hash differs. Its metadata file stores the split sizes, per-epoch history, temperature and every metric reported here. The split is deterministic given the APTOS label file and seed 42. Retraining takes one command (`python -m ai.modules.diabetic_retinopathy.train` with the arguments above, or the provided Kaggle notebook script), and `evaluate.py` re-scores a saved model on any labelled set. The source code, model metadata and test suite are available from the authors on request.

VIII. Regulatory and Ethical Notes

In India, software intended to screen for disease is a medical device under the Medical Devices Rules, 2017. CDSCO's draft guidance on medical device software (October 2025) asks for

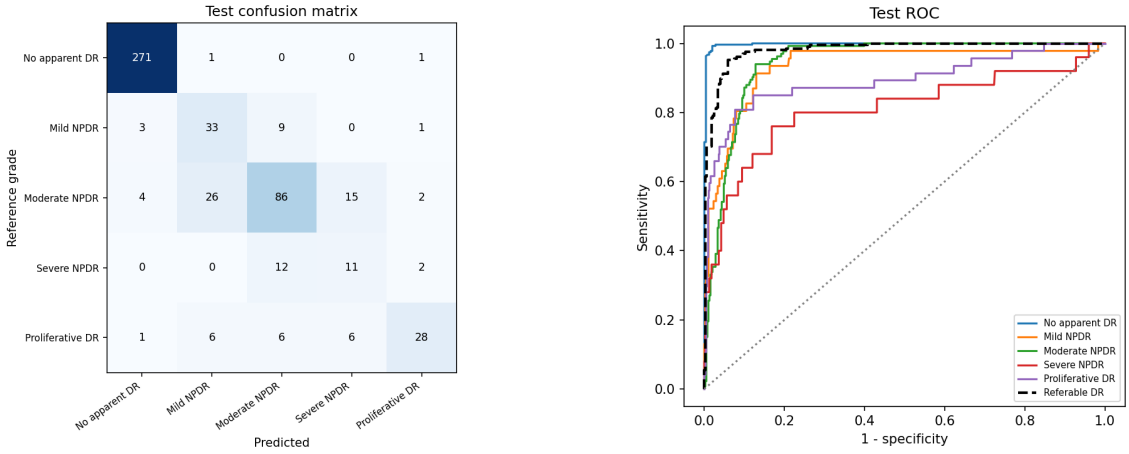


Figure 3: Test set (n = 524): confusion matrix (left; rows are reference grades) and ROC curves (right).

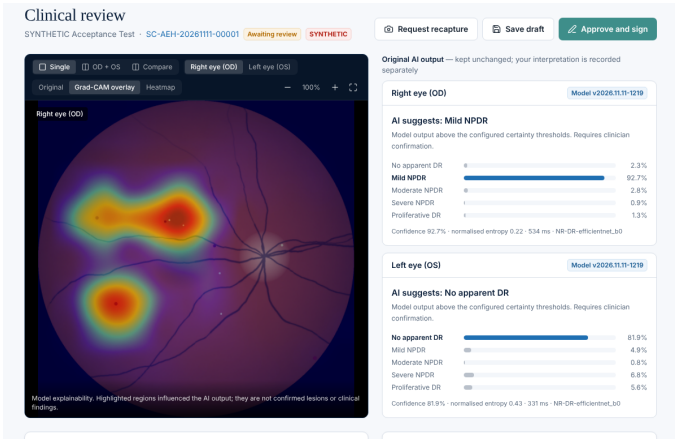


Figure 4: Clinician review screen with a synthetic test patient: Grad-CAM overlay, the model's probabilities, and the per-eye accept/modify/reject controls.

lifecycle quality management, risk management, clinical evaluation and a protocol for changing AI models after approval [15]. We prepared draft intended-use, risk (ISO 14971 [16]) and software documents with IEC 62304 [17] in mind, and consent and erasure follow the Digital Personal Data Protection Act, 2023 [18]. None of this has been assessed by a regulator. All patients and images shown in this paper are synthetic.

IX. Conclusion

An EfficientNet-B0 model trained for under ten minutes on APTOS 2019 reached 81.9% five-grade accuracy and 82.0% / 96.6% sensitivity/specificity for referable DR on a held-out split, and we placed it inside a workflow where it can only suggest. The most important next steps are a per-patient split with duplicates removed, evaluation on images from other hospitals and cameras, tuning the "inconclusive" threshold, and an out-of-distribution check, followed by a prospective study with ethics approval.

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